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CS23532-COMPUTER NETWORKS-LAB MANUAL

PRACTICAL - 7

AIM:

To write and implement a program for flow control at the Data Link Layer using the Sliding Window Protocol, where the sender transmits multiple frames based on a specified window size and the receiver verifies the frames and sends appropriate ACK or NACK acknowledgements.

PROCEDURE:

1. Create the **Sender and Receiver programs** for implementing the Sliding Window Protocol.
2. Enter the **window size** and the text message to be transmitted through the sender program.
3. Divide the message into individual frames and send the frames according to the specified window size.
4. Store the transmitted frames temporarily in Sender_Buffer.txt.
5. The receiver reads the frames, verifies their sequence numbers, and checks whether they are received in the correct order.
6. If all frames are received correctly, the receiver sends an **ACK**; otherwise, it sends a **NACK** for the expected frame.
7. Based on the acknowledgement, the sender slides the window forward or retransmits the required frames until the complete message is successfully transmitted.

PROGRAM CODE:

Sender.c

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>
```

Roll no : 241501199
Name : Sherwin Richie J

```
#ifdef _WIN32
#include <windows.h>

void delay(int seconds)
{
    Sleep(seconds * 1000);
}

#else
#include <unistd.h>

void delay(int seconds)
{
    sleep(seconds);
}

#endif

#define SENDER_FILE "Sender_Buffer.txt"
#define RECEIVER_FILE "Receiver_Buffer.txt"

void clearFile(const char *filename)
{
```

```
    FILE *fp = fopen(filename, "w");

    if (fp != NULL)
        fclose(fp);
}

int readAcknowledgement(int *type, int *number)
{
    FILE *fp;
    char line[100];

    fp = fopen(RECEIVER_FILE, "r");

    if (fp == NULL)
        return 0;

    if (fgets(line, sizeof(line), fp) == NULL)
    {
        fclose(fp);
        return 0;
    }

    fclose(fp);
```

```
    if (sscanf(line, "ACK:%d", number) == 1)
    {
        *type = 1;
        return 1;
    }

    if (sscanf(line, "NACK:%d", number) == 1)
    {
        *type = 2;
        return 1;
    }

    return 0;
}

void writeFrames(char *message, int start, int end)
{
    FILE *fp;

    fp = fopen(SENDER_FILE, "w");

    if (fp == NULL)
    {
        printf("Error: Cannot open Sender_Buffer.txt\n");
```

```
        exit(1);
    }

    printf("\n----------------------------------------\n");
    printf("Sending frames %d to %d\n", start, end);
    printf("----------------------------------------\n");

    for (int i = start; i <= end; i++)
    {
        fprintf(fp, "Frame:%d DATA:%c\n", i, message[i]);

        printf("Frame No: %d DATA: %c\n",
               i, message[i]);
    }

    fclose(fp);
}

int main()
{
    char message[500];
    int windowSize;
    int totalFrames;
    int base = 0;
```

```
int ackType;

int ackNumber;


printf("========== SLIDING WINDOW SENDER ==========\n\n");


printf("Enter window size: ");

scanf("%d", &windowSize);


if (windowSize <= 0)

{

   printf("Invalid window size.\n");

   return 1;

}


getchar();


printf("Enter text message: ");

fgets(message, sizeof(message), stdin);


message[strcspn(message, "\n")] = '\0';


totalFrames = strlen(message);


if (totalFrames == 0)
```

```
{
    printf("Message cannot be empty.\n");
    return 1;
}

clearFile(SENDER_FILE);
clearFile(RECEIVER_FILE);

printf("\nMessage: %s\n", message);
printf("Total frames: %d\n", totalFrames);
printf("Window size: %d\n", windowSize);

while (base < totalFrames)
{
    int end = base + windowSize - 1;

    if (end >= totalFrames)
        end = totalFrames - 1;

    writeFrames(message, base, end);

    printf("\nSender_Buffer.txt updated.");
    printf("\nWaiting for acknowledgement");
```

```
while (1)
{
   delay(2);

   if (readAcknowledgement(&ackType, &ackNumber))
   {
      if (ackType == 1)
      {
         printf("\nReceived ACK:%d\n", ackNumber);

         if (ackNumber >= base)
         {
            base = ackNumber + 1;

            clearFile(RECEIVER_FILE);

            printf("Sliding window moved.\n");
            break;
         }
      }

      else if (ackType == 2)
      {
         printf("\nReceived NACK:%d\n", ackNumber);
```

```
                printf("Resending frames from frame %d...\n",
                    ackNumber);

                clearFile(RECEIVER_FILE);

                if (ackNumber >= base)
                   base = ackNumber;

                break;
            }
        }
    }
}

clearFile(SENDER_FILE);

printf("\n=========================================\n");
printf("All frames transmitted successfully!\n");
printf("Message transmission completed.\n");
printf("=========================================\n");

return 0;
}
```

Roll no : 241501199
Name : Sherwin Richie J

Receiver.c

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>


#ifdef _WIN32

#include <windows.h>


void delay(int seconds)

{

   Sleep(seconds * 1000);

}


#else

#include <unistd.h>


void delay(int seconds)

{

   sleep(seconds);

}


#endif


#define SENDER_FILE "Sender_Buffer.txt"
```

Roll no : 241501199
Name : Sherwin Richie J

```
#define RECEIVER_FILE "Receiver_Buffer.txt"


int readFrames(int *frameNumbers, char *data, int maxFrames)
{
    FILE *fp;
    char line[200];
    int count = 0;

    fp = fopen(SENDER_FILE, "r");

    if (fp == NULL)
        return 0;

    while (fgets(line, sizeof(line), fp) != NULL &&
           count < maxFrames)
    {
        int frameNo;
        char ch;

        if (sscanf(line, "Frame:%d DATA:%c",
                   &frameNo, &ch) == 2)
        {
            frameNumbers[count] = frameNo;
            data[count] = ch;
```

```
            count++;
        }
    }

    fclose(fp);

    return count;
}

void writeACK(int number)
{
    FILE *fp = fopen(RECEIVER_FILE, "w");

    if (fp == NULL)
        return;

    fprintf(fp, "ACK:%d\n", number);

    fclose(fp);
}

void writeNACK(int number)
{
    FILE *fp = fopen(RECEIVER_FILE, "w");
```

```
    if (fp == NULL)
        return;

    fprintf(fp, "NACK:%d\n", number);

    fclose(fp);
}

int main()
{
    int expectedFrame = 0;
    int frameNumbers[100];
    char data[100];
    int lastFirstFrame = -1;

    printf("========== SLIDING WINDOW RECEIVER ==========\n\n");
    printf("Receiver is waiting for frames...\n");

    while (1)
    {
        int count = readFrames(frameNumbers, data, 100);

        if (count > 0)
```

```
{
    if (frameNumbers[0] != lastFirstFrame)
    {
        int valid = 1;

        printf("\n---------------------------------------\n");
        printf("Frames received:\n");
        printf("---------------------------------------\n");

        for (int i = 0; i < count; i++)
        {
            printf("Frame No: %d DATA: %c\n",
                frameNumbers[i],
                data[i]);

            if (frameNumbers[i] != expectedFrame)
            {
                printf("\nExpected Frame: %d\n",
                    expectedFrame);

                printf("Received Frame: %d\n",
                    frameNumbers[i]);

                printf("ERROR DETECTED!\n");
```

```
            printf("Sending NACK:%d\n",
                  expectedFrame);

            writeNACK(expectedFrame);

            valid = 0;
            break;
        }

        expectedFrame++;
    }

    if (valid)
    {
        int ackNumber = expectedFrame - 1;

        printf("\nAll frames received correctly.\n");

        printf("Sending ACK:%d\n",
              ackNumber);

        writeACK(ackNumber);
    }
```

```
            lastFirstFrame = frameNumbers[0];
        }
    }

    delay(1);
  }

  return 0;
}
```

Compilation and Execution

gcc Sender.c -o Sender

gcc Receiver.c -o Receiver

./Receiver

./Sender

Output

Sender Output

========== SLIDING WINDOW SENDER ==========

Enter window size: 3

Enter text message: HELLO

Roll no : 241501199
Name : Sherwin Richie J

Message: HELLO

Total frames: 5

Window size: 3

Sending frames 0 to 2

Frame No: 0 DATA: H

Frame No: 1 DATA: E

Frame No: 2 DATA: L

Received ACK:2

Sliding window moved.

Sending frames 3 to 4

Frame No: 3 DATA: L

Frame No: 4 DATA: O

Received ACK:4

Sliding window moved.

==========

Roll no : 241501199
Name : Sherwin Richie J

All frames transmitted successfully!

Message transmission completed.

==========

Receiver Output

========== SLIDING WINDOW RECEIVER ==========

Receiver is waiting for frames...

Frames received:

Frame No: 0 DATA: H

Frame No: 1 DATA: E

Frame No: 2 DATA: L

All frames received correctly.

Sending ACK:2

Frames received:

Frame No: 3 DATA: L

Frame No: 4 DATA: O

Roll no : 241501199
Name : Sherwin Richie J

All frames received correctly.

Sending ACK:4

RESULT:

Thus, the Sliding Window Protocol was successfully implemented for flow control at the Data Link Layer, where multiple frames were transmitted based on the window size and acknowledgements were used to ensure successful and reliable frame transmission.